Inverse_problem

January 1, 2026

[ ]:
```
from sympy import *

t, a, c = symbols('t a c', real = True)
z = Function('z')(t)
z_d = diff(z, t)
z_dd = diff(z, t, 2)
m = Function('m')(z)
m_dz = diff(m, z)
m_dt = diff(m, t)
```

Defining z's functional form and derivitaves:

[72]:
```
zs = 1/a*(sqrt((a*t)**2+1)-1)
zs
```

[72]:
$$\frac{\sqrt{a^2t^2+1}-1}{a}$$

[73]:
```
zs_d = diff(zs, t)
zs_dd = diff(zs, t, 2)
```

Defining the Lagrangian in code:

[74]:
```
L = -m*sqrt(1-z_d**2)
L
```

[74]:
$$-\sqrt{1-\left(\frac{d}{dt}z(t)\right)^2}\,m(z(t))$$

Taking the partial of the Lagranian with repsect to z:

[75]:
```
diff(L, z)
```

[75]:
$$-\sqrt{1-\left(\frac{d}{dt}z(t)\right)^2}\,\frac{d}{dz(t)}m(z(t))$$

Taking the derivative of L with respect to $\dot{z}$:

[76]:
```
diff(L, z_d)
```

[76]:

$$\frac{m(z(t))\frac{d}{dt}z(t)}{\sqrt{1-\left(\frac{d}{dt}z(t)\right)^2}}$$

Taking the derivative of $\frac{\partial L}{\partial \dot{z}}$ with respect to time:

[77]:
```
diff(diff(L, z_d), t)
```

[77]:

$$\frac{m(z(t))\frac{d^2}{dt^2}z(t)}{\sqrt{1-\left(\frac{d}{dt}z(t)\right)^2}}+\frac{\frac{d}{dz(t)}m(z(t))\left(\frac{d}{dt}z(t)\right)^2}{\sqrt{1-\left(\frac{d}{dt}z(t)\right)^2}}+\frac{m(z(t))\left(\frac{d}{dt}z(t)\right)^2\frac{d^2}{dt^2}z(t)}{\left(1-\left(\frac{d}{dt}z(t)\right)^2\right)^{\frac{3}{2}}}$$

creating EL EQ:

[78]:
```
EL = Eq(diff(L, z), diff(diff(L, z_d), t))
EL
```

[78]:

$$-\sqrt{1-\left(\frac{d}{dt}z(t)\right)^2}\frac{d}{dz(t)}m(z(t)) = \frac{m(z(t))\frac{d^2}{dt^2}z(t)}{\sqrt{1-\left(\frac{d}{dt}z(t)\right)^2}}+\frac{\frac{d}{dz(t)}m(z(t))\left(\frac{d}{dt}z(t)\right)^2}{\sqrt{1-\left(\frac{d}{dt}z(t)\right)^2}}+\frac{m(z(t))\left(\frac{d}{dt}z(t)\right)^2\frac{d^2}{dt^2}z(t)}{\left(1-\left(\frac{d}{dt}z(t)\right)^2\right)^{\frac{3}{2}}}$$

Solving the Euler-Lagrange Eq for $\frac{\partial m}{\partial z}$:

[79]:
```
m_ode = Eq(m_dz,solve(EL,m_dz)[0])
m_ode
```

[79]:

$$\frac{d}{dz(t)}m(z(t)) = \frac{m(z(t))\frac{d^2}{dt^2}z(t)}{\left(\frac{d}{dt}z(t)\right)^2-1}$$

Pulling m to the left side & Splitting this into two integrals:

[80]:
```
m_ode.subs(m,1).rhs
```

[80]:

$$\frac{\frac{d^2}{dt^2}z(t)}{\left(\frac{d}{dt}z(t)\right)^2-1}$$

Noting that:

$$\int f(z(t))dz = \int f(z(t))\frac{dz}{dt}dt$$

[81]:
```
ode_sol = Eq(ln(m),(integrate((Eq(m**-1*m_dz,solve(EL,m_dz)[0].subs(m,1)).rhs.
↪subs(z,zs)*z_d).subs(z,zs).doit(), t))+c)
ode_sol
```

[81]:

$$\log(m(z(t))) = c-\frac{\log(a^2t^2+1)}{2}$$

[82]:
```
solve(ode_sol,m)[0]
```

[82]:

$$\frac{e^c}{\sqrt{a^2t^2+1}}$$

solving z for $\sqrt{a^2t^2+1}$ gives:

$$az+1=\sqrt{a^2t^2+1}$$

this implies

$$m(t)=\frac{e^c}{\sqrt{a^2t^2+1}} \rightarrow m(z)=\frac{e^c}{az+1}$$

Plugging back into the Lagrangian:

$$L=-\frac{e^c}{az+1}\sqrt{1-\left(\frac{d}{dt}z(t)\right)^2}$$

Comparing to interview question Lagrangian:

$$L=-\frac{m_0}{az+1}\sqrt{1-\left(\frac{d}{dt}z(t)\right)^2}$$